FP-Devicer: Open-Source Digital Fingerprinting Middleware

Technical Whitepaper — Version 1.6.0+

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1 Abstract

FP-Devicer is a lightweight, highly extensible TypeScript middleware library for server-side digital device fingerprinting. It computes a **confidence score (0–100)** between two fingerprint datasets by combining **weighted structural field-by-field comparison** with **TLSH (Trend Micro Locality Sensitive Hash) fuzzy holistic scoring**.

The library provides a simple `calculateConfidence` API for one-off comparisons and a full-featured `DeviceManager` paired with a pluggable `StorageAdapter` system for production-grade device identification, snapshot persistence, deduplication, adaptive weighting based on historical signal stability, and structured observability.

Key design goals:

- Near-universal server compatibility (Express, Fastify, Deno/Oak, etc.)
- Extensibility via a global plugin registry (custom weights and comparators per field path)
- High accuracy (>99% in benchmarks) with sub-millisecond per-comparison latency
- Multiple storage backends: in-memory, SQLite, PostgreSQL, and Redis

The codebase (~2,500 lines of clean TypeScript) is modular, well-tested, benchmarked, and documented via TypeDoc. It pairs naturally with the companion client-side collector **FP-Snatch** for a complete end-to-end fingerprinting solution.

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2 Introduction & Motivation

Traditional fingerprinting libraries (e.g., FingerprintJS open-source) focus on client-side collection and hash generation. FP-Devicer shifts the intelligence to the **server side**, enabling capabilities that are impossible or impractical in the browser:

- **Persistent device IDs** that survive across browser sessions, private mode tabs, and even across different browsers on the same device
- **Adaptive scoring** that automatically down-weights signals that have proven historically volatile for a given device (e.g., screen orientation, timezone)
- **Candidate pre-filtering** to keep database queries fast even at scale — only broadly-similar fingerprints are loaded for full scoring
- **Enterprise features** such as user account linking, IP address logging, and structured metrics emission

FP-Devicer was released open-source in 2025 and has seen active development (93+ commits), with the most recent patch on March 8, 2026. It is available on GitHub at https://github.com/gatewaycorporate/fp-devicer and its generated TypeDoc reference documentation is hosted at https://gatewaycorporate.github.io/fp-devicer/.

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3 Architecture Overview

FP-Devicer is structured into five logical layers, each with a clearly defined responsibility:

Data Model (`src/types/data.ts`) : Defines `FPUserDataSet`, the canonical interface for all fingerprint signals collected from a browser, and the generic `FPDataSet<T>` alias. Also contains configuration types such as `ComparisonOptions` and `Comparator`.

Scoring Engine (`src/libs/confidence.ts` + `src/libs/registry.ts`) : Implements the two-strategy hybrid confidence calculation: recursive structural scoring and TLSH fuzzy hashing. The registry is a global singleton that stores per-path weights and comparator functions contributed by built-in defaults or user plugins.

Persistence & Orchestration (`src/core/DeviceManager.ts` + `src/libs/adapters/`) : The `DeviceManager` class runs the full fingerprint matching pipeline. Storage adapters implement a common interface so that the engine code is entirely decoupled from the choice of database.

Plugin Architecture (`src/core/PluginRegistrar.ts`) : The `PluginRegistrar` manages `DeviceManagerPlugin` objects that hook into the post-identification pipeline. Each plugin implements a single `registerWith()` method that receives a `DeviceManagerLike` handle and registers one or more `IdentifyPostProcessor` callbacks. First-party companion libraries (`ip-devicer`, `tls-devicer`) use this mechanism to enrich `IdentifyResult` with IP intelligence and TLS consistency data without coupling to FP-Devicer internals.

Observability (`src/types/observability.ts` + `src/libs/default-observability.ts`) : Defines injectable `Logger` and `Metrics` interfaces with no-op defaults, allowing callers to route telemetry to any logging or metrics platform.

The relationships among components are summarized in the class diagram below.

```
┌──────────────────────────────────────────────────────────────────┐
│                          FP-Devicer                              │
│                                                                  │
│  FPDataSet ──► ConfidenceCalculator ◄── Registry                 │
│                        ▲                                         │
│                        │                                         │
│  StorageAdapter ──► DeviceManager ◄─── PluginRegistrar           │
│        ▲               │                      ▲                  │
│  (SQLite / Postgres    │               DeviceManagerPlugin       │
│   Redis / InMemory)    └──► IdentifyPostProcessors               │
│                                   ▲                              │
```

```
│                          (ip-devicer IpManager,                  │
│                           tls-devicer TlsManager, …)             │
└──────────────────────────────────────────────────────────────────┘
```

---

4 Core Components

4.1 Fingerprint Data Model

The `FPUserDataSet` interface (defined in `src/types/data.ts`) is the canonical schema for all browser signals that FP-Snatch collects client-side and FP-Devicer consumes server-side. It contains more than 30 optional fields, allowing partial fingerprints to be processed gracefully. Relevant fields include:

Field	Type	Description
`userAgent`	`string`	Full browser user-agent string
`platform`	`string`	OS platform string (e.g., `"Win32"`)
`screen`	`object`	Width, height, color depth, orientation
`fonts`	`string[]`	Enumerated installed system fonts
`canvas`	`string`	Hash of a rendered canvas element
`webgl`	`string`	Hash of WebGL rendering output
`audio`	`string`	Hash of an AudioContext fingerprint
`plugins`	`object[]`	Browser plugin names and descriptions
`mimeTypes`	`object[]`	Supported MIME types
`hardwareConcurrency`	`number`	Logical CPU core count
`deviceMemory`	`number`	Reported device RAM (GiB, rounded)
`highEntropyValues`	`object`	UA-CH high-entropy hints (brands, model, etc.)
`language` / `languages`	`string` / `string[]`	Browser locale information
`timezone`	`string`	IANA timezone identifier

All fields are optional. The scoring engine treats a field that is absent on one side of the comparison as a zero-similarity match for that path rather than throwing an error.

The generic `FPDataSet<T>` alias allows the scoring engine to operate on shapes other than `FPUserDataSet`, supporting advanced use cases where callers supply custom data models:

```
export type FPDataSet<T = FPUserDataSet> = T;
```

4.2 Confidence Scoring Engine

The scoring engine lives in `src/libs/confidence.ts` and is the intellectual core of FP-Devicer. It exposes two entry points:

- `calculateConfidence` — a pre-built default calculator, exported directly for simple use cases.
- `createConfidenceCalculator(options)` — a factory that returns a calculator instance scoped to a specific `ComparisonOptions` configuration.

4.2.1 Strategy 1: Weighted Structural Comparison

The structural comparison is performed by `compareRecursive`, an internal function that walks both fingerprint objects simultaneously using dot-notation paths. At each node it determines whether the value is:

- **Primitive / leaf** — delegates to the registered comparator for that path (or the built-in default) and returns a similarity in `[0, 1]`.
- **Array** — pairs elements by index and recurses into each pair, accumulating weighted contributions from each element.
- **Object** — takes the union of keys present in either object and recurses into each key pair.

Weights are retrieved via `getWeight(path)`, which consults the merged weight table built from `userOptions.weights`, the global registry, and built-in `DEFAULT_WEIGHTS`. The weights are **normalized internally**, meaning only relative magnitudes matter; it is not necessary for them to sum to any particular value.

The structural score is computed as:

$$S_{\text{structural}} = \frac{\sum_i w_i \cdot \text{sim}_i}{\sum_i w_i}$$

where w_i is the effective weight for path i and $\text{sim}_i \in [0, 1]$ is the similarity returned by the path's comparator.

4.2.2 Strategy 2: TLSH Fuzzy Hashing

TLSH (Trend Micro Locality Sensitive Hash) is a fuzzy hashing algorithm designed so that similar inputs produce similar hashes. Unlike cryptographic hashes, two TLSH hashes can be compared to produce a numeric distance score that correlates with the semantic similarity of the original inputs.

FP-Devicer serializes each fingerprint to a canonical JSON string (key-sorted, deterministic) before hashing. The hash distance is normalized to a `[0, 1]` similarity score:

$$S_{\text{tlsh}} = 1 - \frac{\text{distance}\ (h_1, h_2)}{\text{maxDistance}}$$

TLSH serves as a holistic cross-check. Because it operates on the serialized whole rather than individual fields, it catches fingerprint-wide shifts that the structural scorer might partially miss when individual fields receive low weight.

4.2.3 Score Blending

The two strategies are combined by a configurable `tlshWeight` parameter (default `0.30`):

$$S_{\text{final}} = S_{\text{structural}} \cdot (1 - w_{\text{tlsh}}) + S_{\text{tlsh}} \cdot w_{\text{tlsh}}$$

The final result is scaled to the integer range `[0, 100]` and returned. A score of `100` indicates an exact or near-exact match; a score of `0` indicates no detectable similarity.

4.2.4 Options Resolution Order

When `createConfidenceCalculator` builds its internal weight and comparator tables it merges sources in the following priority order (highest first):

1. `userOptions.weights` / `userOptions.comparators` (caller-supplied overrides)
2. Built-in `DEFAULT_WEIGHTS` (hardcoded high-entropy emphasis)
3. Global registry entries added via `registerPlugin` / `registerWeight`
4. `defaultWeight` fallback (default `1`)

4.2.5 ComparisonOptions Reference

```
interface ComparisonOptions {
    /** Per-path weights. Normalized automatically. */
    weights?: Record<string, number>;
    /** Per-path custom similarity functions. */
    comparators?: Record<string, Comparator>;
    /** Fallback weight for paths without an explicit entry. Default: 1 */
    defaultWeight?: number;
    /** TLSH blend factor in [0, 1]. Default: 0.30 */
    tlshWeight?: number;
    /** Maximum object recursion depth. Default: 8 */
    maxDepth?: number;
    /** Whether to consult the global registry. Default: true */
    useGlobalRegistry?: boolean;
}
```

4.3 Plugin & Registry System

The registry (`src/libs/registry.ts`) is a **global singleton** that stores per-path comparators and weights. It is lazily seeded by `initializeDefaultRegistry()` (called from `src/libs/default-plugins.ts`) the first time any scoring function runs.

Built-in defaults include:

- **Jaccard similarity** for set-like arrays (`fonts`, `languages`, `plugins`, `mimeTypes`) — measures the ratio of the intersection size to the union size.
- **Exact match** for hash fields (`canvas`, `webgl`, `audio`) — returns `1.0` only when both values are identical strings.

- **`screenSimilarity`** for the `screen` object — a custom function that tolerates small pixel-rounding differences while penalizing large resolution changes.
- **Structural equality** as a fallback for primitives with no registered comparator.

Callers can extend or override the registry at any time using the exported helper functions:

```
// Register both a custom weight and a custom comparator at once
registerPlugin("userAgent", {
    weight: 25,
    comparator: (a, b) =>
        levenshteinSimilarity(
            String(a ?? "").toLowerCase(),
            String(b ?? "").toLowerCase(),
        ),
});

// Register only a weight (keep the existing comparator)
registerWeight("canvas", 40);

// Register only a comparator (keep the existing weight)
registerComparator("timezone", (a, b) => (a === b ? 1 : 0.5));

// Set the fallback weight for any unregistered path
setDefaultWeight(2);

// Remove a registration
unregisterWeight("audio");
unregisterComparator("audio");
```

Registry changes take effect immediately for all subsequent calls to `calculateConfidence` (including via `DeviceManager`) because the global registry is consulted at scoring time, not at calculator-creation time.

4.4 Device Management

`DeviceManager` (`src/core/DeviceManager.ts`) is the high-level orchestrator that makes FP-Devicer suitable for production use. It wraps the scoring engine with a complete fingerprint matching pipeline:

4.4.1 Pipeline Steps

1. **Deduplication cache** — Computes the TLSH hash of the incoming fingerprint and checks an in-memory LRU cache (keyed by hash, 5-second TTL). If the same fingerprint is seen twice within the window, the cached `IdentifyResult` is returned immediately without any database I/O. This is critical for endpoints that receive burst traffic.

2. **Candidate pre-filtering** — Calls `adapter.findCandidates(incoming, minScore)` to retrieve a small set of stored device snapshots that are broadly similar to the incoming data. Each adapter implements this differently: the SQLite and PostgreSQL adapters

use JSON-operator `WHERE` clauses on indexed fields; the in-memory adapter does a linear scan with early termination. The default `candidateMinScore` is `30`, meaning only candidates where the adapter's rough score exceeds 30 are returned.

3. **Adaptive weighting** — For each candidate, `DeviceManager` calls `adapter.getHistory(deviceId, limit=5)` to retrieve recent snapshots and passes them to `computeFieldStabilities`. This function measures, for each fingerprint path, how consistent the field's value has been across the historical snapshots. Fields that have varied frequently (e.g., `timezone` for a travelling user) are assigned a reduced weight for this comparison, preventing transient signal drift from causing a false negative.
4. **Full confidence scoring** — Each candidate is scored against its most recent snapshot using a `createConfidenceCalculator` instance configured with the adaptive weights computed in step 3. The candidate with the highest score is selected as `bestMatch`.
5. **Decision** — If `bestMatch.confidence >= matchThreshold` (default `50`), its existing device ID is reused. Otherwise a new UUID-based device ID is minted, marking the visit as a new device.
6. **Persistence** — The incoming fingerprint is saved via `adapter.save()` as a new snapshot associated with the resolved device ID.
7. **Observability** — Structured log entries are emitted via the injected `Logger`. Metrics counters and gauges are incremented via the injected `Metrics` instance. Both default to no-op implementations.
8. **Post-processing** — Each registered `IdentifyPostProcessor` is called in registration order with the full pipeline outcome. Processors may merge additional fields onto the result, attach structured enrichment data under `enrichmentInfo.details[name]`, and emit additional log metadata. Processor failures are caught individually — a throwing processor records an entry in `enrichmentInfo.failures` and execution continues with the next processor. See §4.4 for the full plugin architecture.

4.4.2 Constructor Options

```
new DeviceManager(adapter, {
  matchThreshold?: number;      // min score to reuse a device ID (default: 50)
  candidateMinScore?: number;   // pre-filter floor passed to adapter (default:
30)
  dedupWindowMs?: number;       // dedup cache TTL in milliseconds (default:
5000)
  comparisonOptions?: ComparisonOptions; // forwarded to
createConfidenceCalculator
  observability?: {
    logger?: Logger;
    metrics?: Metrics;
  };
});
```

4.4.3 Return Value

```
interface IdentifyResult {
    deviceId: string; // stable UUID for this device
    confidence: number; // 0–100 score against the best matched snapshot
    isNewDevice: boolean; // true if no candidate exceeded matchThreshold
    matchConfidence: number; // mirror of confidence; persisted on the snapshot
    linkedUserId?: string; // if a userId was supplied and stored
    enrichmentInfo: {
        plugins: string[]; // names of processors that ran
        details: Record<string, Record<string, unknown>>; // per-plugin
structured data
        failures: Array<{ plugin: string; message: string }>; // caught processor
errors
    };
}
```

4.5 Universal Plugin Architecture for Device Managers

FP-Devicer provides a second, distinct extension mechanism that operates at the **pipeline level** rather than the field level. Where the scoring registry (§4.2) customizes *how individual fingerprint fields are compared*, the plugin architecture customizes *what happens after a device has been identified*. The two systems are orthogonal and can be combined freely.

4.5.1 Design Overview

The architecture is built from three collaborating types:

- **`DeviceManagerPlugin`** (`src/core/PluginRegistrar.ts`) — the interface all plugins implement. It declares a single method, `registerWith`, that receives a `DeviceManagerLike` handle and uses it to subscribe to pipeline hooks.
- **`PluginRegistrar`** (`src/core/PluginRegistrar.ts`) — a class owned by `DeviceManager` that validates and tracks active plugins. It is exported publicly so custom orchestrators can host plugins against any compatible manager.
- **`DeviceManagerLike`** (`src/core/DeviceManager.ts`) — a minimal interface that exposes only the methods plugins need. Decoupling this from the concrete `DeviceManager` class prevents circular imports and makes plugins testable against lightweight fakes.

4.5.2 `DeviceManagerPlugin` Interface

```
interface DeviceManagerPlugin {
    /**
     * Called immediately when the plugin is registered via DeviceManager.use().
     * May return an optional teardown function invoked on unregister.
     */
    registerWith(deviceManager: DeviceManagerLike): (() => void) | void;
}
```

Plugins are registered by calling `DeviceManager.use(plugin)`, which internally delegates to `PluginRegistrar.register()`. That method validates that `registerWith` is a function

(throwing `"Invalid plugin: Missing 'registerWith' method."` otherwise), calls it with the manager, and returns an unregister function that calls any returned teardown and removes the plugin from the active list.

```
const unregister = manager.use(myPlugin);

// Later, to cleanly tear down:
unregister();
```

The currently active plugins are inspectable at any time:

```
const active: readonly DeviceManagerPlugin[] = manager.getPlugins();
```

4.5.3 `DeviceManagerLike` Interface

```
interface DeviceManagerLike {
    registerIdentifyPostProcessor(
        name: string,
        processor: IdentifyPostProcessor,
    ): () => void;
}
```

This minimal surface is the only handle exposed to plugin code, preventing plugins from accidentally mutating manager state outside the sanctioned hook. `registerIdentifyPostProcessor` accepts a unique `name` (used as the key in `enrichmentInfo.details`) and a processor function, and returns an unregister closure.

4.5.4 `IdentifyPostProcessor` Lifecycle

After the core pipeline (dedup → candidates → score → decide → save) completes, `DeviceManager` invokes each registered post-processor in registration order. Each receives an `IdentifyPostProcessorPayload` and may return an `IdentifyPostProcessorResult` (or nothing).

`IdentifyPostProcessorPayload`

Field	Type	Description
`incoming`	`FPDataSet`	The raw fingerprint from the current request
`context`	`IdentifyContext?`	Per-request context passed to `identify()` (e.g., `ip`, `userId`, `tlsProfile`)
`result`	`IdentifyResult`	The result as enriched by all preceding processors
`baseResult`	`IdentifyResult`	A fresh clone of the core result, unaffected by other processors
`cacheHit`	`boolean`	Whether the dedup cache was used for this request
`candidatesCount`	`number`	Number of candidates returned by `findCandidates()`
`matched`	`boolean`	Whether an existing device was matched
`durationMs`	`number`	Elapsed time of the core pipeline in milliseconds

`IdentifyPostProcessorResult`

Field	Type	Description
`result`	`Record<string, unknown>?`	Fields merged directly onto `IdentifyResult` (e.g., custom scores)
`enrichmentInfo`	`Record<string, unknown>?`	Structured data stored under `enrichmentInfo.details[name]`
`logMeta`	`Record<string, unknown>?`	Extra key-value pairs bundled into the structured log entry

4.5.5 `IdentifyEnrichmentInfo`

Every `IdentifyResult` carries an `enrichmentInfo` object that accumulates the contributions of all post-processors for that request:

Field	Type	Description
`plugins`	`string[]`	Names of all processors that ran
`details`	`Record<string, Record<string, unknown>>`	Per-plugin structured enrichment data
`failures`	`Array<{ plugin: string; message: string }>`	Errors caught from failed processors

4.5.6 Error Isolation

If a processor throws or rejects, the error is caught, a `{ plugin, message }` entry is appended to `enrichmentInfo.failures`, and a `logger.warn` line is emitted. Execution continues with the next registered processor. No processor failure can prevent the core `IdentifyResult` from being returned to the caller.

4.5.7 Companion Plugin: ip-devicer `IpManager`

`IpManager` from the **ip-devicer** companion package implements `DeviceManagerPlugin`. When registered, it hooks into the post-processor pipeline to perform real-time IP intelligence enrichment on every `identify()` call.

Registration:

```
import { IpManager } from "ip-devicer";
import { createSqliteAdapter, DeviceManager } from "devicer.js";

const ipManager = new IpManager({/* MaxMind DB paths, thresholds */});
const manager = new DeviceManager(createSqliteAdapter("./fp.db"), {
    matchThreshold: 60,
});
await manager.adapter.init();

manager.use(ipManager);
```

Enriched fields added to `IdentifyResult`:

Field	Type	Description
`ipEnrichment`	`object`	Country, ASN, agent info, risk score, consistency score, impossible-travel flag, and proxy/VPN/Tor/hosting flags
`ipRiskDelta`	`number`	Change in risk score since the device's previous visit

The processor reads the resolved IP from `context.ip` (or `context.resolvedIp` as set by `createIpMiddleware`). If no IP is present in the context, the processor returns without enriching the result.

4.5.8 Companion Plugin: tls-devicer `TlsManager`

`TlsManager` from the **tls-devicer** companion package implements `DeviceManagerPlugin`. It cross-references the TLS/JA4 fingerprint attached to the request against stored TLS profiles for the matched device, producing a consistency score and optionally boosting the confidence value.

Registration:

```
import { createTlsMiddleware, TlsManager } from "tls-devicer";
import { createSqliteAdapter, DeviceManager } from "devicer.js";
```

```
const tlsManager = new TlsManager({/* options */});
const manager = new DeviceManager(createSqliteAdapter("./fp.db"), {
    matchThreshold: 60,
});
await manager.adapter.init();

manager.use(tlsManager);
```

The `createTlsMiddleware()` helper extracts JA4/JA3 hashes, HTTP/2 SETTINGS, and header-order signals from each request and attaches them as `req.tlsProfile`. Pass this into `identify()` via the context:

```
app.post("/identify", async (req, res) => {
    const result = await manager.identify(req.body, {
        tlsProfile: req.tlsProfile,
    });
    res.json(result);
});
```

Enriched fields added to `IdentifyResult`:

Field	Type	Description
`tlsConsistency`	`object`	Consistency score (0–1), JA4/JA3 match flags, and per-factor breakdown
`tlsConfidenceBoost`	`number`	Signed delta applied to `result.confidence` based on TLS consistency
`confidence`	`number`	Overwritten with the TLS-boosted value (original preserved in `matchConfidence`)

4.6 Storage Adapters

All adapters implement the `StorageAdapter` interface defined in `src/types/storage.ts`. The interface methods are:

Method	Description
`init()`	Create tables / open connections. Must be called before use.
`save(snapshot)`	Persist a new fingerprint snapshot.
`findCandidates(fp, n)`	Return up to `n` broadly-similar stored snapshots.
`getHistory(id, limit)`	Return recent snapshots for a given device ID.
`getAllFingerprints()`	Return all stored snapshots (e.g., for bulk analysis).
`linkToUser(id, uid)`	Associate a device ID with an application user ID.
`deleteOldSnapshots()`	Prune snapshots older than a configured retention window.
`close()`	(Optional) Gracefully close database connections.

The four provided adapter implementations handle the specifics:

In-memory (`createInMemoryAdapter()`) : Stores snapshots in a `Map`. Suitable for development, testing, and short-lived server processes. No persistence across restarts.

SQLite (`createSqliteAdapter(path)`) : Uses `better-sqlite3` (or the Deno-compatible equivalent). Creates a `fingerprints` table on first `init()`. The candidate pre-filter runs a `WHERE` clause that compares stored platform and screen values before loading the full rows. Ideal for single-process servers without external dependencies.

PostgreSQL (`createPostgresAdapter(connectionString)`) : Uses Drizzle ORM (`drizzle-orm/postgres-js`). Suitable for multi-process or horizontally-scaled deployments. The candidate pre-filter leverages PostgreSQL JSON operators for efficient indexed querying.

Redis (`createRedisAdapter(url)`) : Stores snapshots as JSON-serialized Redis hashes. Suitable for deployments that already operate a Redis cluster or that require very fast writes-per-second with eventual persistence.

The `AdapterFactory` class provides a configuration-driven factory for selecting and instantiating adapters without importing each adapter module directly:

```
const adapter = AdapterFactory.create("sqlite", {
    sqlite: { path: "./fingerprints.db" },
});
```

---

5 Code Paths & Sequence Diagrams

5.1 Simple Confidence Calculation

The simplest use case requires no instance management:

```
Client
  │
  ├─► calculateConfidence(fp1, fp2)
  │       │
  │       ├─► initializeDefaultRegistry()  [lazy, once only]
  │       │       │
  │       │       └─► registerPlugin × N   [canvas, fonts, screen, ...]
  │       │
  │       ├─► compareRecursive(fp1, fp2)
  │       │       │
  │       │       └─► for each field path:
  │       │               getWeight(path)
  │       │               getComparator(path)(val1, val2)
  │       │
  │       ├─► getHash(canonicalizedStringify(fp1))
  │       ├─► getHash(canonicalizedStringify(fp2))
  │       └─► compareHashes(h1, h2)  → tlsh score
  │
  └◄─ blended integer score [0–100]
```

5.2 Full Device Identification

```
HTTP Request
  │
  ├─► DeviceManager.identify(incomingFP, { userId, ip })
  │       │
  │       ├─► getHash(incoming)  →  dedupKey
  │       ├─► dedupCache.get(dedupKey)
  │       │       ├─ HIT  → return cached IdentifyResult immediately
  │       │       └─ MISS → continue
  │       │
  │       ├─► adapter.findCandidates(incoming, candidateMinScore)
  │       │       └─► returns [ candidate₁, candidate₂, ... ]
  │       │
  │       ├─► for each candidate:
  │       │       ├─► adapter.getHistory(candidate.deviceId, 5)
  │       │       ├─► computeFieldStabilities(history)
  │       │       ├─► createConfidenceCalculator({ weights: adaptiveWeights })
  │       │       └─► calculator.calculateConfidence(incoming, latestSnapshot)
  │       │
  │       ├─► select bestMatch (highest score)
  │       │
  │       ├─► if bestMatch.score >= matchThreshold:
  │       │       deviceId = bestMatch.deviceId   (returning device)
  │       │   else:
  │       │       deviceId = crypto.randomUUID()  (new device)
  │       │
  │       ├─► adapter.save({ deviceId, fingerprint: incoming, ... })
  │       ├─► dedupCache.set(dedupKey, baseResult, dedupWindowMs)
  │       │
```

```
  │       ├─► for each identifyPostProcessor [plugin]:
  │       │       ├─► processor({ incoming, context, result, baseResult,
  │       │       │                cacheHit, candidatesCount, matched,
durationMs })
  │       │       │       └─► returns { result?, enrichmentInfo?, logMeta? }
  │       │       ├─► merge returned result fields → enrichedResult
  │       │       ├─► append enrichedResult.enrichmentInfo.details[plugin]
  │       │       └─► on throw: append enrichmentInfo.failures[plugin], continue
  │       │
  │       ├─► logger.info({ ...perPluginLogMeta })
  │       └─► metrics.incrementCounter(...)
  │
  └─◄ IdentifyResult { deviceId, confidence, isNewDevice, enrichmentInfo, ... }
```

6 Extensibility & Customization

6.1 Custom Comparators

A `Comparator` is any function with the signature:

```
type Comparator = (value1: any, value2: any, path?: string) => number;
```

It must return a value in `[0, 1]` where `1.0` means identical and `0.0` means completely dissimilar. The optional `path` argument allows a single comparator to behave differently depending on which field it is invoked for.

Common comparator patterns include:

- **Levenshtein similarity** for string fields that change incrementally (e.g., user-agent minor version bumps).
- **Jaccard similarity** for set-valued fields (arrays where order does not matter).
- **Numeric proximity** for quantitative fields (e.g., `deviceMemory` treated as similar if within one doubling step).
- **Exact match** for high-entropy hash fields where any difference is significant.

6.2 Custom Storage Adapters

To integrate FP-Devicer with a database not covered by the built-in adapters, implement the `StorageAdapter` interface:

```
import { FPDataSet, StorageAdapter, StoredFingerprint } from "devicer.js";

const myAdapter: StorageAdapter = {
    async init() {/* open connection, create schema */},
    async save(snapshot: StoredFingerprint) {/* insert row */},
    async findCandidates(fp: FPDataSet, minScore: number, limit: number) {
        /* return StoredFingerprint[] of broadly-similar records */
        return [];
    },
```

```
    async getHistory(deviceId: string, limit: number) {
        return [];
    },
    async getAllFingerprints() {
        return [];
    },
    async linkToUser(deviceId: string, userId: string) {},
    async deleteOldSnapshots(olderThanMs: number) {},
    async close() {/* close connection */},
};
```

6.3 Custom Observability

Inject a `Logger` and/or `Metrics` implementation into `DeviceManager` to route telemetry to your platform (e.g., Winston, Pino, StatsD, Prometheus):

```
import { createInMemoryAdapter, DeviceManager } from "devicer.js";

const manager = new DeviceManager(createInMemoryAdapter(), {
    observability: {
        logger: {
            info: (msg, meta) => pinoLogger.info(meta, msg),
            warn: (msg, meta) => pinoLogger.warn(meta, msg),
            error: (msg, meta) => pinoLogger.error(meta, msg),
        },
        metrics: {
            incrementCounter: (name, value = 1) => statsd.increment(name, value),
            recordGauge: (name, value) => statsd.gauge(name, value),
            recordHistogram: (name, value) => statsd.histogram(name, value),
        },
    },
});
```

6.4 Device Manager Plugins

To integrate custom post-identification logic with any `DeviceManager`, implement the `DeviceManagerPlugin` interface exported from `devicer.js`:

```
import type {
    DeviceManagerLike,
    DeviceManagerPlugin,
    IdentifyPostProcessorPayload,
} from "devicer.js";

const riskPlugin: DeviceManagerPlugin = {
    registerWith(deviceManager: DeviceManagerLike) {
        const unregister = deviceManager.registerIdentifyPostProcessor(
            "risk-scorer",
            async ({ result, context }: IdentifyPostProcessorPayload) => {
                const riskScore = await myRiskApi.score(result.deviceId);
                return {
```

```
                    result: { riskScore },
                    enrichmentInfo: { riskScore, deviceId: result.deviceId },
                    logMeta: { riskScore },
                };
            },
        );
        // Return the unregister closure as the plugin teardown function
        return unregister;
    },
};

const unregisterRisk = manager.use(riskPlugin);

// To remove the plugin cleanly:
unregisterRisk();
```

Any fields returned in `result` are merged onto `IdentifyResult` and immediately visible to processors registered after this one. The teardown function (the unregister closure from `registerIdentifyPostProcessor`) is called automatically when `unregisterRisk()` is invoked, so no manual cleanup is required.

---

7 Performance & Accuracy

7.1 Benchmarking Methodology

FP-Devicer ships with a rigorous, self-contained benchmark suite under `src/benchmarks/`. All benchmarks run under the Vitest framework using its built-in `bench` harness, which applies statistical analysis across many iterations to produce stable, reproducible results. The suite is split into two concerns: **accuracy** (does the scorer make correct decisions?) and **performance** (how fast does it run?).

7.1.1 Synthetic Dataset Generation

All accuracy benchmarks operate on synthetic data produced by `data-generator.ts`. The generator is fully deterministic — it uses a seeded Linear Congruential Generator (LCG) as its PRNG so that results are reproducible across environments and CI runs. No external data sources or network calls are required.

For each simulated device a *base fingerprint* is constructed by sampling from realistic value pools:

- **Platform & OS** — 11 OS/architecture combinations covering Windows 10/11, macOS Intel/ARM, Ubuntu, Android 13/14, and iOS 16/17.
- **Browser** — 14 browser profiles (Chrome, Firefox, Safari, Edge, Opera across desktop and mobile) with versioned user-agent templates; version numbers are sampled from a realistic recency window.

- **Locale & timezone** — draws from a pool of 45 IANA timezone identifiers paired with culturally appropriate language lists.
- **Screen** — width × height × colour depth combinations sampled from 10 common display resolutions.
- **Hardware entropy** — `hardwareConcurrency` and `deviceMemory` sampled from common real-world values; UA-CH high-entropy hints populated accordingly.
- **Hash-based signals** — `canvas`, `webgl`, and `audio` fingerprints are generated by dedicated functions (`generateCanvasBlob`, `generateWebGLBlob`, `generateAudioBlob`) that produce multi-component strings mirroring the output of FP-Snatch's collection routines. Each function has a *stable* portion (seeded deterministically from the device seed, representing GPU/hardware baseline) and a *jitter* portion (varied per call, representing per-render floating-point noise). The raw strings are then passed through the same djb2 hash used by FP-Snatch.
- **Font list** — drawn from a pool of real system fonts; count and order vary per device.

7.1.1.1 Attractor Devices

Approximately 10–15% of simulated devices are designated **attractor devices**. These are generated by `createAttractorFingerprint` and represent common hardware/software profiles (e.g., default Windows 10 Chrome with a typical screen resolution) that produce fingerprints densely packed in feature space. Attractor devices are the hardest case for a fingerprint scorer because two completely different users may share nearly identical feature vectors. The `isAttractor` flag is propagated to every pair in the accuracy evaluation, and the `attr` column in the output table reports the False Accept Rate computed exclusively over impostor pairs where at least one party is an attractor device.

7.1.2 Mutation Model

Each device is assigned `sessionsPerDevice` fingerprint snapshots by applying the `mutate()` function at escalating intensity levels, cycling through `none → low → medium → high → low`. On 40% of non-`none` sessions, a second `low`-intensity pass is layered on top to ensure no two samples from the same device are ever bit-for-bit identical:

Level	Signals Modified
`none`	Clean baseline — no changes from the generated base
`low`	±1–2 px screen width rounding; canvas hash micro-jitter; 40% chance of font list reorder
`medium`	Chrome minor version bump; font addition or removal; 20% chance of timezone change; canvas and audio re-hash
`high`	Screen resolution change; Chrome major version jump (+2–6); DoNotTrack toggle; multiple font changes; canvas and WebGL hash drift from simulated driver update
`extreme`	Entirely new device profile generated from a fresh seed — treated as a different device for scoring purposes

The `extreme` level is excluded from the standard accuracy test corpus because it represents a hardware replacement, not a re-visit.

7.1.3 Pair Construction

The accuracy benchmark (`accuracy.bench.ts`) generates **5,750 scored pairs** from a corpus of 2,000 devices × 5 sessions each:

- **Genuine pairs** (`sameDevice = true`): For each iteration `i`, two distinct sessions from the same device are selected and passed to `calculateConfidence`. These represent legitimate re-visits that should score above the decision threshold.
- **Random impostor pairs** (`sameDevice = false`): A session from a separate device is paired with one from the target device. These represent different users that should score below the threshold.
- **Cross-browser impostor pairs** (`sameDevice = false`, deliberately harder): For 30% of iterations, a cross-browser pair is constructed by pairing a session from the target device with one sampled from an attractor-zone device. This tests resistance to the most difficult false-accept scenario.

All pair scores are computed synchronously at module load time before the Vitest bench harness starts timing, so DBSCAN analysis ($O(n^2)$) does not pollute the performance measurement.

7.1.4 Accuracy Metrics

Pair scores are evaluated across 21 thresholds from 0 to 100 in steps of
5. For each threshold the following metrics are computed:

Metric	Formula	Interpretation
Precision	$\mathrm{TP}\ /(\mathrm{TP}+\mathrm{FP})$	Of all pairs accepted, what fraction were genuine?
Recall	$\mathrm{TP}\ /(\mathrm{TP}+\mathrm{FN})$	Of all genuine pairs, what fraction were correctly accepted?
F1	$2 \cdot P \cdot R/(P+R)$	Harmonic mean of precision and recall
FAR	$\mathrm{FP}\ /(\mathrm{FP}+\mathrm{TN})$	Rate at which impostors are incorrectly accepted
FRR	$\mathrm{FN}\ /(\mathrm{TP}+\mathrm{FN})$	Rate at which genuine pairs are incorrectly rejected
EER	$\mid \mathrm{FAR}-\mathrm{FRR} \mid$	Proximity of FAR and FRR — minimum indicates the crossover point
attr	FAR over attractor-only impostor pairs	Hardened FAR for the most challenging impostor category

The threshold with the highest F1 score is selected as the recommended operating point. Results are persisted to `src/benchmarks/benchmark.out` after each run.

A complementary **DBSCAN analysis** (`eps = 0.05`, `minPts = 3`) is run over the full scored-pair distribution to characterise separability. DBSCAN groups pairs with similar scores into clusters; a large number of tight, distinct clusters indicates good bimodal separation between genuine and impostor score populations. The summary reports cluster count, noise points (pairs with anomalous scores not belonging to any cluster), total clustered points, and the largest cluster size.

7.1.5 Performance Benchmarks

`performance.bench.ts` measures end-to-end latency across the three deployment configurations most likely to be used in production:

Benchmark	What is timed
`calculateConfidence` (hybrid scorer)	Pure scoring only — structural comparison + TLSH blend, no I/O
`DeviceManager.identify` (In-Memory)	Full pipeline: dedup cache check, candidate scan, adaptive weighting, scoring, save
`DeviceManager.identify` (SQLite in-memory)	Same pipeline with SQLite `:memory:` as the storage backend
`DeviceManager.identify` (SQLite file-based)	Same pipeline with a real on-disk SQLite database, reflecting realistic conditions

Each adapter is seeded with 5 warmup `identify` calls before timing begins to ensure the dedup cache, query plan cache, and JIT compiler are in a steady state. The harness runs for 6 seconds per configuration (`time: 6000`) with a minimum of 50 iterations.

7.2 Benchmark Results

Results are from the most recent run against the current codebase on an 8-core x86_64 development machine. Timings are in milliseconds.

7.2.1 Accuracy (2,000 devices, 5,750 pairs)

Metric	Value
Best threshold	60
Precision at threshold 60	0.988
Recall at threshold 60	0.991
F1 at threshold 60	0.989
EER at threshold 60	0.000
FAR at threshold 60	0.010
FRR at threshold 60	0.009
Attractor impostor FAR (attr)	0.041
DBSCAN clusters	76
DBSCAN noise points	8 / 5750

The zero EER at threshold 60 is the Pareto-optimal operating point where FAR and FRR cross. The attractor FAR of 4.1% at this threshold represents the primary remaining risk: devices with common hardware profiles whose fingerprints cluster densely together can occasionally produce false accepts. This rate drops to zero above threshold 70, but at that point FRR rises to 12.6%, making threshold 60 the practical recommendation.

7.2.2 Performance

Configuration	Mean (ms)	p75 (ms)	p99 (ms)	Throughput (ops/sec)	RME
`calculateConfidence` (scorer only)	1.13	1.12	1.63	884	±1.74%
`DeviceManager` (In-Memory)	0.77	0.75	1.69	1,301	±0.52%
`DeviceManager` (SQLite in-memory)	0.78	0.78	1.54	1,274	±0.52%
`DeviceManager` (SQLite file-based)	0.76	0.75	1.32	1,321	±0.84%

The `DeviceManager` configurations outperform the raw scorer in mean latency because the dedup cache short-circuits repeated fingerprints before any scoring occurs. At steady-state traffic where a meaningful fraction of requests are repeat device visits, the effective throughput is substantially higher than the worst-case figures above.

SQLite file-based latency is comparable to in-memory at p75 and p99 due to SQLite's write-ahead logging and OS page cache effects; the higher tail at p999 (1.62 ms vs 2.29 ms for in-memory) reflects occasional fsync overhead.

To reproduce these results locally:

```
npm run bench
```

Results are written to `src/benchmarks/bench-results.json` (raw Vitest output) and `src/benchmarks/benchmark.out` (formatted accuracy tables).

---

8 Testing

FP-Devicer ships with a comprehensive, multi-layered test suite located under `src/tests/`. All tests run under Vitest and are organized into four distinct categories: **unit tests**, **integration tests**, **storage contract tests**, and **fixture-driven regression tests**. Together they cover every public API surface, internal library function, storage adapter implementation, and cross-cutting resilience concern.

8.1 Test Organization

```
src/tests/
├── data-generator.test.ts        # Synthetic data generator validation
├── fixtures/
│   └── fingerprints.ts           # Shared deterministic fingerprint fixtures
├── core/
│   ├── adapter-factory.test.ts   # AdapterFactory configuration-driven creation
│   ├── device-manager.test.ts    # DeviceManager pipeline & decision logic
│   └── plugin-registrar.test.ts  # PluginRegistrar contract & lifecycle
```

```
├── libs/
│   ├── comparitors.test.ts       # Built-in comparator functions
│   ├── confidence.test.ts        # Scoring engine & blending logic
│   ├── registry.test.ts          # Plugin registry CRUD and precedence
│   └── tlsh.test.ts              # TLSH hashing & distance normalization
├── storage/
│   ├── adapter-contract.test.ts  # Shared behavioural contract for all adapters
│   ├── inmemory-adapter.test.ts  # In-memory adapter specifics
│   ├── postgres-adapter.test.ts  # PostgreSQL adapter specifics
│   ├── redis-adapter.test.ts     # Redis adapter specifics
│   └── sqlite-adapter.test.ts    # SQLite adapter specifics
└── integration/
    ├── api-surface.test.ts       # Public API contract & ergonomics
    └── resilience.test.ts        # Fault tolerance & edge-case handling
```

8.2 Fixtures

`fixtures/fingerprints.ts` provides a set of deterministic, hand-authored `FPUserDataSet` objects shared across all unit and integration tests. Using static fixtures ensures that any regression in scoring output is immediately visible — a field change in a fixture produces a reproducible, diff-able failure rather than a flaky random result. Fixtures cover:

- A canonical **baseline** fingerprint representing a typical desktop Chrome visit.
- **Similar variant** fingerprints with minor mutations (screen rounding, font reorder, canvas micro-jitter) to verify that genuine-pair scores remain high.
- **Dissimilar** fingerprints drawn from completely different OS/browser profiles to verify that impostor-pair scores remain low.
- **Partial / sparse** fingerprints with large numbers of `undefined` fields to verify graceful degradation.
- **Edge-case** fingerprints containing empty arrays, null-like values, and maximum-entropy hash strings.

8.3 Unit Tests

8.3.1 Comparator Functions (`libs/comparitors.test.ts`)

Tests every built-in comparator exported from `src/libs/comparitors.ts` in isolation:

- **Jaccard similarity** — verifies set intersection / union ratios for font and plugin arrays, including empty-set handling (expected result: `0`) and identical-set handling (expected result: `1`).
- **Exact match** — verifies `1.0` for identical strings and `0.0` for any difference, including case sensitivity.
- **Screen similarity** — verifies tolerance for small pixel-rounding differences (±1–2 px) while producing low scores for resolution changes, and correct handling of missing `orientation` fields.
- **Numeric proximity** — verifies that values within a configured step return near-`1` scores and that out-of-range values degrade gracefully.

Each comparator is exercised with boundary inputs (both values `undefined`, one value `undefined`, empty string, zero) to confirm the absence of thrown exceptions.

8.3.2 TLSH Hashing (`libs/tlsh.test.ts`)

Tests the `getHash` and `compareHashes` functions from `src/libs/tlsh.ts`:

- Verifies that two calls to `getHash` with identical inputs produce identical hash strings (determinism).
- Verifies that `compareHashes` returns `1.0` when both hashes are identical and a value strictly less than `1.0` for different hashes.
- Verifies that the normalized distance is always in `[0, 1]` regardless of input content or length.
- Verifies that `getHash` applied to a canonical JSON serialization produces the same result as applying it to an equivalent object serialized independently, confirming key-order independence.

8.3.3 Plugin Registry (`libs/registry.test.ts`)

Tests the global registry singleton via its full public API:

- `registerPlugin`, `registerWeight`, `registerComparator` — verifies that registered values are immediately retrievable via `getWeight` and `getComparator`.
- `unregisterWeight`, `unregisterComparator` — verifies that removal causes the affected path to fall back to the default weight / comparator.
- `setDefaultWeight` — verifies that unregistered paths return the updated default.
- **Isolation** — each test resets the registry to a clean state before running to prevent inter-test pollution.
- **Precedence** — verifies the documented resolution order: caller `userOptions` override registry entries, which override built-in defaults.

8.3.4 Device Manager (`core/device-manager.test.ts`)

Tests the full `DeviceManager` pipeline and decision logic:

- **Dedup cache** — a second call within `dedupWindowMs` returns the cached result without a DB write.
- **New-device minting** — when no candidate exceeds `matchThreshold`, a fresh UUID is returned with `isNewDevice: true`.
- **Returning-device matching** — a stored fingerprint exceeding the threshold returns the existing `deviceId` with `isNewDevice: false`.
- **Adaptive weighting** — verifies that a field with high historical variance is down-weighted in subsequent comparisons.
- **`enrichmentInfo` structure** — every `IdentifyResult` carries an `enrichmentInfo` object with `plugins`, `details`, and `failures` arrays initialized to empty values before any post-processors run.

- **Post-processor results** — a registered `IdentifyPostProcessor` can merge new fields onto the result and attach structured data under `enrichmentInfo.details[name]`.
- **Post-processor failure isolation** — a processor that throws appends an entry to `enrichmentInfo.failures` and does not prevent subsequent processors from running.
- **Post-processor sequencing** — multiple processors are called in registration order; each sees the `result` as mutated by all preceding processors.
- **`use()` / `getPlugins()` / unregister** — `manager.use(plugin)` delegates to `PluginRegistrar`, the plugin appears in `getPlugins()`, and the returned unregister function calls the plugin's teardown and removes it from the active list.
- **Plugin `enrichmentInfo` end-to-end** — a plugin registered via `use()` that returns `enrichmentInfo` has its data visible in the final `IdentifyResult`.

8.3.5 PluginRegistrar (`core/plugin-registrar.test.ts`)

Tests the `PluginRegistrar` class in isolation via a lightweight `DeviceManagerLike` fake:

- **`registerWith` invocation** — `register()` calls `plugin.registerWith` with the provided manager instance exactly once.
- **Invalid plugin validation** — passing an object whose `registerWith` is not a function throws `"Invalid plugin: Missing 'registerWith' method."`
- **Teardown on unregister** — the returned unregister function calls the teardown returned by `registerWith`.
- **Active plugin enumeration** — `getRegisteredPlugins()` returns the active set; after unregistering, the plugin no longer appears.
- **Multiple plugins** — registering two plugins and unregistering one leaves the other active.

8.3.6 Confidence Scoring (`libs/confidence.test.ts`)

Tests `calculateConfidence` and `createConfidenceCalculator` against the shared fixtures:

- **Identical inputs** produce a score of `100`.
- **Completely unrelated inputs** produce a score at or near `0`.
- **Similar variants** (minor mutations) produce scores above the recommended operating threshold of `60`.
- **Custom weight overrides** shift scores in the expected direction (increasing the weight of a mutated field lowers the score; decreasing it raises the score).
- **`tlshWeight: 0`** disables TLSH blending and produces a score driven entirely by structural comparison, verified by comparing against a manually computed expected value.
- **`tlshWeight: 1`** produces a score driven entirely by TLSH, verified similarly.
- **`maxDepth`** is respected — recursion is capped and does not throw for deeply nested objects.
- **`useGlobalRegistry: false`** causes registry entries to be ignored, verified by registering a custom comparator that would otherwise affect the score.

8.3.7 Synthetic Data Generator (`data-generator.test.ts`)

Tests `generateFingerprint`, `mutate`, and `createAttractorFingerprint` from `src/benchmarks/data-generator.ts`:

- Verifies that two calls with the same seed produce bit-for-bit identical fingerprints (LCG determinism).
- Verifies that two calls with different seeds produce different fingerprints.
- Verifies that `mutate` at level `none` returns a fingerprint equal to the base.
- Verifies that `mutate` at levels `low` through `high` returns a fingerprint that differs from the base in at least one field but shares enough structure to score above `60` when compared against it.
- Verifies that `createAttractorFingerprint` sets the `isAttractor` flag and produces a fingerprint within the expected common-hardware value pools.

8.4 Storage Tests

8.4.1 Shared Adapter Contract (`storage/adapter-contract.test.ts`)

A single parameterized test suite is executed against every adapter implementation. This approach ensures that all adapters exhibit identical observable behaviour regardless of their underlying persistence mechanism. The contract tests cover:

- **`init()`** — adapter can be initialized without errors; calling `init()` twice is idempotent.
- **`save()` + `getAllFingerprints()`** — a saved snapshot is retrievable and its fields match what was passed to `save`.
- **`findCandidates()`** — a saved fingerprint is returned as a candidate when queried with a sufficiently similar fingerprint; an entirely dissimilar fingerprint is not returned.
- **`getHistory()`** — multiple saves for the same `deviceId` are all returned by `getHistory`, ordered with the most recent first, and the result is limited to the requested count.
- **`linkToUser()`** — after linking, `getHistory` returns snapshots that carry the associated `userId`.
- **`deleteOldSnapshots()`** — snapshots older than the supplied retention window are removed; newer snapshots are retained.
- **`close()`** — completes without throwing when the adapter has been initialized and when it has not.

8.4.2 Per-Adapter Specialization Tests

Each adapter's dedicated test file extends the shared contract with backend-specific concerns:

In-Memory (`inmemory-adapter.test.ts`) : Verifies that the `Map`-backed store is empty on construction, that data does not persist after the adapter object is garbage-collected, and that concurrent `save` calls do not corrupt the internal state.

SQLite (`sqlite-adapter.test.ts`) : Verifies schema creation on first `init()`, that the `fingerprints` table uses the expected column layout, that the candidate pre-filter `WHERE`

clause correctly limits returned rows, and that the `:memory:` path and a real file path both work correctly.

PostgreSQL (`postgres-adapter.test.ts`) : Verifies that the Drizzle ORM schema migration runs cleanly, that JSON-operator candidate queries return results consistent with the contract expectations, and that the connection is properly released on `close()`. These tests require a live PostgreSQL instance and are skipped automatically when the `TEST_POSTGRES_URL` environment variable is absent.

Redis (`redis-adapter.test.ts`) : Verifies that snapshots are serialized to and deserialized from Redis hashes without data loss, that TTL-based expiry is configured correctly, and that `close()` disconnects the client. These tests require a live Redis instance and are skipped automatically when `TEST_REDIS_URL` is absent.

8.5 Integration Tests

8.5.1 API Surface (`integration/api-surface.test.ts`)

Exercises the full public API as a consumer would use it, without mocking any internal modules:

- `calculateConfidence` is importable and callable with no configuration.
- `createConfidenceCalculator` returns a stateful calculator that produces consistent results across multiple calls.
- `DeviceManager` constructed with an in-memory adapter completes a full `identify` round-trip and returns a well-formed `IdentifyResult`.
- A device re-identified with the same fingerprint within `dedupWindowMs` returns a cached result with `isNewDevice: false` and the same `deviceId`.
- A fingerprint that does not match any stored candidate returns `isNewDevice: true` and mints a new UUID.
- `AdapterFactory.create` resolves the correct adapter class for each supported type string (`"inmemory"`, `"sqlite"`, `"postgres"`, `"redis"`).
- All exported types (`FPUserDataSet`, `StorageAdapter`, `IdentifyResult`, `ComparisonOptions`, etc.) are present on the module namespace.

8.5.2 Resilience (`integration/resilience.test.ts`)

Verifies fault-tolerance and edge-case handling throughout the pipeline:

- **Empty fingerprint objects** (`{}`) do not throw; they return a score of `0` from `calculateConfidence` and are assigned a new device ID by `DeviceManager`.
- **`null` / `undefined` field values** within an otherwise valid fingerprint do not throw and are treated as absent fields.
- **Adapter `findCandidates` returning an empty array** causes `DeviceManager` to mint a new device ID rather than throwing.
- **Adapter `save` throwing** is caught and surfaced as a structured error log entry via the injected `Logger` rather than an unhandled rejection.

- **Deeply nested objects** beyond `maxDepth` are truncated rather than triggering a stack overflow.
- **Very large fingerprints** (hundreds of fields) complete in under the sub-millisecond latency budget verified by the performance benchmarks.
- **Concurrent `identify` calls** for the same fingerprint hash within the dedup window all return the same `deviceId` without race conditions.
- **Registry state isolation** — modifications made to the global registry during one test do not leak into subsequent tests.

8.6 Running the Test Suite

```
# Run all tests once
npm test

# Run in watch mode during development
npm run test:watch

# Run with coverage report
npm run test:coverage
```

Storage adapter tests that require external services use environment variables to configure connection strings. When the variables are absent the tests are skipped cleanly rather than failing:

```
# Run with PostgreSQL and Redis adapters included
TEST_POSTGRES_URL=postgresql://localhost/fp_test \
TEST_REDIS_URL=redis://localhost:6379 \
npm test
```

---

9 Usage Examples

9.1 Method 1: Simple (Using Defaults)

```
import { calculateConfidence } from "devicer.js";

const score = calculateConfidence(fpData1, fpData2);
// score is an integer in [0, 100]
```

9.2 Method 2: Advanced (Custom Weights & Comparators)

```
import { createConfidenceCalculator, registerPlugin } from "devicer.js";

// Contribute a globally-scoped plugin before creating any calculators
registerPlugin("userAgent", {
    weight: 25,
    comparator: (a, b) =>
        levenshteinSimilarity(
            String(a ?? "").toLowerCase(),
            String(b ?? "").toLowerCase(),
```

```
        ),
});

const calculator = createConfidenceCalculator({
    weights: {
        platform: 20,
        fonts: 20,
        screen: 15,
    },
    tlshWeight: 0.2,
});

const score = calculator.calculateConfidence(fpData1, fpData2);
```

9.3 Method 3: Enterprise (DeviceManager with Express)

```
import express from "express";
import { createInMemoryAdapter, DeviceManager } from "devicer.js";

const manager = new DeviceManager(createInMemoryAdapter(), {
    matchThreshold: 60,
    candidateMinScore: 30,
});
await manager.adapter.init();

const app = express();
app.use(express.json());

app.post("/identify", async (req, res) => {
    const result = await manager.identify(req.body, {
        userId: (req as any).user?.id,
        ip: req.ip,
    });
    res.json(result);
    // → { deviceId, confidence, isNewDevice, linkedUserId }
});

app.listen(
    3000,
    () => console.log("FP-Devicer server ready at http://localhost:3000"),
);
```

9.4 Method 4: Deno / Oak (FP-Cicis Pattern)

```
import { Application, Router } from "oak";
import { DeviceManager } from "devicer";
import { createSqliteAdapter } from "./libs/sqlite.ts";

const adapter = createSqliteAdapter("./fp.db");
await adapter.init();
```

```
const manager = new DeviceManager(adapter, {
    matchThreshold: 60,
    candidateMinScore: 40,
});

const router = new Router();
router.post("/identify", async (ctx) => {
    const body = await ctx.request.body.json();
    const result = await manager.identify(body);
    ctx.response.body = result;
});
```

9.5 Method 5: Universal Plugins (ip-devicer + tls-devicer)

```
import express from "express";
import { createSqliteAdapter, DeviceManager } from "devicer.js";
import { createIpMiddleware, IpManager } from "ip-devicer";
import { createTlsMiddleware, TlsManager } from "tls-devicer";

// Construct the manager and companion plugins
const adapter = createSqliteAdapter("./fp.db");
const manager = new DeviceManager(adapter, { matchThreshold: 60 });
const ipManager = new IpManager({/* MaxMind DB options */});
const tlsManager = new TlsManager();

await adapter.init();

// Register companion plugins — order determines post-processor execution order
manager.use(tlsManager); // runs first: may boost result.confidence via JA4 match
manager.use(ipManager); // runs second: adds IP intelligence to the result

const app = express();
app.use(express.json());
app.use(createTlsMiddleware()); // attaches req.tlsProfile on every request

app.post("/identify", createIpMiddleware(), async (req: any, res) => {
    const result = await manager.identify(req.body, {
        userId: req.user?.id,
        ip: req.resolvedIp, // set by createIpMiddleware
        tlsProfile: req.tlsProfile, // set by createTlsMiddleware
    });

    console.log(result.deviceId); // stable device UUID
    console.log(result.confidence); // base score ± TLS confidence boost
    console.log(result.tlsConsistency); // { consistencyScore, ja4Match, factors,
… }
    console.log(result.ipEnrichment); // { country, asn, riskScore, isVpn, … }
    console.log(result.enrichmentInfo); // { plugins: ["tls-devicer","ip-
devicer"], details, failures }
```

```
    res.json(result);
});

app.listen(3000);
```

---

10 Installation & Integration

10.1 Installation

```
npm install devicer.js
```

10.2 FP-Snatch Integration

FP-Devicer is designed to pair with **FP-Snatch**, a companion open-source client-side JavaScript library that collects all fields of `FPUserDataSet` from a visitor's browser and POSTs them as JSON to a server endpoint.

```
<script src="./dist/bundle.js"></script>
<script defer>
    var agent = new window["snatch"]({
        url: "https://myserver.example/identify",
        method: "POST",
    });
    agent.send();
</script>
```

FP-Snatch collects, among other signals: user-agent, platform, screen dimensions, installed fonts, canvas fingerprint, WebGL fingerprint, audio fingerprint, plugin list, MIME types, hardware concurrency, device memory, timezone, language preferences, and UA-CH high-entropy values. It then sends the dataset as a JSON body matching the `FPUserDataSet` schema, ready for direct consumption by `DeviceManager.identify()` or `calculateConfidence()`.

A full working demonstration of the FP-Snatch → FP-Devicer pipeline is publicly hosted at cicis.info.

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11 Conclusion & Future Work

FP-Devicer delivers production-grade, open-source device intelligence with unmatched extensibility and accuracy for its class. Its hybrid structural-plus-TLSH scoring, adaptive per-device stability weighting, multi-backend persistence options, and zero-dependency observability interfaces set it apart from simpler fingerprint hash libraries.

The library is suitable for a wide range of use cases: account security and fraud detection, analytics de-duplication, personalisation without cookies, and research into browser fingerprinting techniques.

Community contributions are welcome. Possible future directions include:

- Official npm publishing and semantic versioning automation
- Additional storage adapters (DynamoDB, MongoDB, Cloudflare KV)
- Machine-learning-assisted comparator parameter tuning
- Browser extension support for richer signal collection
- Differential privacy mechanisms for privacy-preserving fingerprinting

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12 License

See `license.txt` in the repository root. FP-Devicer is released under a permissive open-source license.